

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 37

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) • Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas • Small scale matches local demand; electricity used locally without long transmission lines / grid connection • Low construction cost and simple technology maintained by the community • Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed; Kaplan turbines with adjustable blades cope with variable flow	5
01.2	• Lack of capital / finance, technical knowledge or grid access; seasonal drying of rivers; conflicts with irrigation water; planning / land ownership	1
01.3	• Output is proportional to flow x efficiency: $100 \times 0.10 \times (40 / 90)$ • = 4.4 kW (accept 4.4-4.5)	2
01.4	<i>Any 2 from:</i> • Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost; high flows that clear gravel spawning beds lost; floodplain no longer flooded • Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis • Sediment trapped so less nutrient replenishment downstream; barrier to migrating fish (salmon, sturgeon) and isolation of populations	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
02.1	• High-level: used fuel rods / fission products separated in reprocessing; vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) • Intermediate: fuel cladding / metal tubes around fuel rods, filters from reprocessing; mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults • Low-level solid: contaminated clothing, equipment; sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg)	6
02.2	• Radioactive decay of the concentrated fission products generates heat which could damage containers • Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years	2
02.3	• Liquids: filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) • Gases: filtered (HEPA / charcoal filters) to remove particles and iodine before release	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	
03.1	• The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence • An individual event, or even several, does not show a trend; the change may be in the frequency of events which can only be assessed years later • Changes occur at different time and spatial scales (local storm vs global trend); interconnected systems make attribution complex	3
03.2	• Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) • As the town grows around the weather station the warming recorded is partly local, not global; older equipment / sparse global coverage also reduce reliability	2
03.3	• Tree rings: width of each annual ring shows the growth rate, indicating temperature / rainfall in that year; rings easily dated by counting; limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions • Ice cores: annual snow layers compact into ice; air bubbles trapped in the ice give the CO₂ (and methane) concentration when trapped; oxygen isotope ratios indicate temperature; up to 800 000 years (Dome C); limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers; only polar regions • Others accepted: coral growth rings; pollen in lake sediments dated by radiocarbon	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	
04.1	• $(68\,000 - 7\,000) / 68\,000 \times 100$ • = 89.7% (accept 90%)	2
04.2	• Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume	2

Question	Marking guidance	Marks
04.3	- Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake - Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) - Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) - Loss of livelihoods; conflict between the countries sharing the rivers	4
04.4	- Upstream abstraction for irrigation reduces the flow available downstream - Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
05.1	B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
05.2	- Chemicals cause the oil molecules to join together / solidify - Producing more solid material that can be collected / lifted more easily than liquid oil	2
05.3	- Air contains about 210 cm³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm³ per litre (less than 3% of the amount in air) - So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	
06.1	1 nitrogen fixation; 2 nitrification 3 denitrification; 4 ammonification 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three)	3
06.2	- Drainage makes the soil aerobic - Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate - Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas	3
06.3	- Nitrogen gas is very unreactive so most organisms cannot use it directly - Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount	2
06.4	- Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins - When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
07.1	- The human ear is sensitive to a very wide range of sound pressures / powers (about 10¹² range) - A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity	2

Question	Marking guidance	Marks
07.2	• Difference = 30 dB = 3 x 10 dB • = 10^3 = 1000 times	2
07.3	• 88 dB • Doubling the sound intensity increases the level by 3 dB (logarithmic scale)	2
07.4	• 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing; quieter sounds exist but cannot be heard	1
07.5	• 20 to 20 000 Hz • dB(A) weights sounds between about 1000 and 4000 Hz more heavily • Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance; range narrows with age or noise damage	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	
08.1	• $(6400 - 220) / 6400 \times 100$ • = 96.6% (accept 97%)	2
08.2	<i>Any 3 from:</i> • Switch from coal to natural gas (which is desulfurised) / closure of coal-fired power stations removed the main SO₂ source • Flue gas desulfurisation fitted to remaining coal plants; coal / oil desulfurisation • NO_x comes mainly from road transport and vehicle numbers / distance travelled increased greatly • NO_x is produced from air (nitrogen + oxygen) in any high-temperature combustion, so cannot be removed from the fuel like sulfur • Catalytic converters only work when warm / diesel engines produce more NO_x / older vehicles remained in use	3
08.3	• Ozone and PANs are secondary pollutants formed by photochemical reactions involving NO₂ (broken down by UV to monatomic oxygen which forms ozone) • If the primary pollutant is reduced the reactants are not present so the secondary pollutants cannot form	2
08.4	• Lichens are very sensitive to acidic conditions / SO₂ and different species have different tolerances • The species present, their abundance, size / health can be used as a biotic index of the pollution level • They give a long-term integrated measure of pollution rather than a single instantaneous reading; cheap, no equipment needed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-253; AQA spec 3.4.3.2.1-3.4.3.2.3.	
09.1	• The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean • The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly	2
09.2	• t-test • Comparing two means of measured (continuous) data	2
09.3	• There is a 30% probability that the difference occurred by random chance • Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted	2

Question	Marking guidance	Marks
09.4	- Two correlated variables may both be caused by a third variable that was not investigated - Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation	2
09.5	- Increase the number of replicates (reduces the effect of variability) - Standardise other variables (soil, water, light) more closely to reduce scatter	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel - Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade - Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses - Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) - Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants - Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) - Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) - Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together - Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring - Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9

Question	Marking guidance	Marks
10.2	The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
11.1	Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Properties: energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts - Fossil fuels: choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports; new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) - Nuclear: high energy density reduces transport; waste properties (half-life) determine storage; thorium and fusion prospects - Renewables: matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting); intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity; low-impact designs (helical turbines, VAWT, tidal lagoons) - Energy conservation reduces demand: transport, buildings, industry (CHP, heat exchangers) - Impacts before and during use; life-cycle / embodied energy - Evaluation: no resource is impact-free; knowledge allows the best mix and mitigation; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25